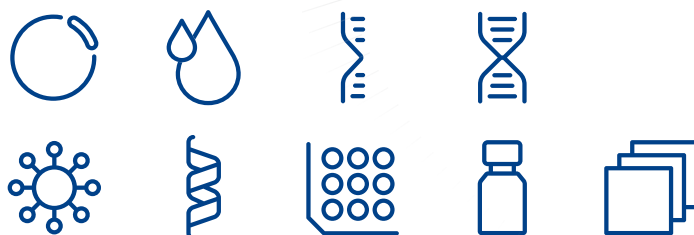


MACHEREY-NAGEL

User manual



Plasmid DNA purification

- NucleoBond® Xtra Midi
- NucleoBond® Xtra Maxi
- NucleoBond® Xtra Midi Plus
- NucleoBond® Xtra Maxi Plus

February 2021 / Rev. 16

Plasmid DNA purification (NucleoBond® Xtra Midi / Maxi)

Protocol at a glance (Rev. 16)

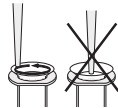
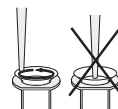
		Midi	Maxi	
1–3 Cultivation and harvest	4,500–6,000 x g 4 °C, 15 min			
4–5 Cell lysis <i>(Important: Check Buffer LYS for precipitated SDS)</i>	High-copy / low-copy 8 mL / 16 mL Buffer RES 8 mL / 16 mL Buffer LYS RT, 5 min		High-copy / low-copy 12 mL / 24 mL Buffer RES 12 mL / 24 mL Buffer LYS RT, 5 min	
6 Equilibration of the column and filter	12 mL Buffer EQU  25 mL Buffer EQU			
7 Neutralization	8 mL / 16 mL Buffer NEU Mix thoroughly until colorless		12 mL / 24 mL Buffer NEU Mix thoroughly until colorless	
8 Clarification and loading of the lysate	Invert the tube 3 times Load lysate on NucleoBond® Xtra Column Filter			
9 1 st Wash	5 mL Buffer EQU		15 mL Buffer EQU	
10 Filter removal	Discard NucleoBond® Xtra Column Filter		Discard NucleoBond® Xtra Column Filter	
11 2 nd Wash	8 mL Buffer WASH		25 mL Buffer WASH	
12 Elution	5 mL Buffer ELU		15 mL Buffer ELU	
13 Precipitation	NucleoBond® Xtra Midi	NucleoBond® Xtra Midi Plus	NucleoBond® Xtra Maxi	NucleoBond® Xtra Maxi Plus
	3.5 mL Isopropanol Vortex 4,5–15,000 x g 4 °C, 30 min	3.5 mL Isopropanol Vortex RT, 2 min Load NucleoBond® Finalizer	10.5 mL Isopropanol Vortex 4,5–15,000 x g 4 °C, 30 min	10.5 mL Isopropanol Vortex RT, 2 min Load NucleoBond® Finalizer Large
14 Washing and drying	2 mL 70 % ethanol 4,5–15,000 x g RT, 5 min 10–15 min	2 mL 70 % ethanol ≥ 6 x air until dry	4 mL 70 % ethanol 4,5–15,000 x g RT, 5 min 15–30 min	4 mL 70 % ethanol ≥ 6 x air until dry
15 Reconstitution	Appropriate volume of TE buffer	200–800 µL Buffer TRIS	Appropriate volume of TE buffer	400–1000 µL Buffer TRIS



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1 Components

1.1 Kit contents

	NucleoBond® Xtra Midi			NucleoBond® Xtra Midi Plus	
REF	10 preps 740410.10	50 preps 740410.50	100 preps 740410.100	10 preps 740412.10	50 preps 740412.50
Buffer RES	100 mL	500 mL	1000 mL	100 mL	500 mL
Buffer LYS	100 mL	500 mL	1000 mL	100 mL	500 mL
Buffer NEU	100 mL	500 mL	1000 mL	100 mL	500 mL
Buffer EQU	200 mL	1000 mL	2 x 1000 mL	200 mL	1000 mL
Buffer WASH	100 mL	500 mL	1000 mL	100 mL	500 mL
Buffer ELU	60 mL	300 mL	600 mL	60 mL	300 mL
RNase A* (lyophilized)	6 mg	30 mg	60 mg	6 mg	30 mg
NucleoBond® Xtra Midi Columns incl. NucleoBond® Xtra Midi Column Filters	10	50	100	10	50
NucleoBond® Finalizers	-	-	-	10	50
30 mL Syringes	-	-	-	10	50
1 mL Syringes	-	-	-	10	50
Buffer TRIS	-	-	-	13 mL	60 mL
Plastic Washers (reusable)	5	10	10	5	10
User manual	1	1	1	1	1

* For preparation of working solutions and storage conditions see section 5.

1.1 Kit contents *continued*

REF	NucleoBond® Xtra Maxi			NucleoBond® Xtra Maxi Plus	
	10 preps 740414.10	50 preps 740414.50	100 preps 740414.100	10 preps 740416.10	50 preps 740416.50
Buffer RES	150 mL	750 mL	2 x 750 mL	150 mL	750 mL
Buffer LYS	150 mL	750 mL	2 x 750 mL	150 mL	750 mL
Buffer NEU	150 mL	750 mL	2 x 750 mL	150 mL	750 mL
Buffer EQU	500 mL	2 x 1000 mL 500 mL	5 x 1000 mL	500 mL	2 x 1000 mL 500 mL
Buffer WASH	300 mL	1000 mL 500 mL	3 x 1000 mL	300 mL	1000 mL 500 mL
Buffer ELU	180 mL	900 mL	2 x 900 mL	180 mL	900 mL
RNase A* (lyophilized)	10 mg	50 mg	2 x 50 mg	10 mg	50 mg
NucleoBond® Xtra Maxi Columns incl. NucleoBond® Xtra Maxi Column Filters	10	50	100	10	50
NucleoBond® Finalizers Large	-	-	-	10	50
30 mL Syringes	-	-	-	10	50
1 mL Syringes	-	-	-	10	50
Buffer TRIS	-	-	-	13 mL	60 mL
Plastic Washers (reusable)	5	10	10	5	10
User manual	1	1	1	1	1

* For preparation of working solutions and storage conditions see section 5.

1.2 Reagents and equipment to be supplied by user

Reagents

- Isopropanol (room-temperated)
- 70 % ethanol (room-temperated)
- Buffer for reconstitution of DNA, for example TE buffer or sterile H₂O (not necessary for NucleoBond® Xtra Midi/Maxi Plus kits)

Equipment

Standard microbiological equipment for growing and harvesting bacteria (e.g., inoculating loop, culture tubes and flasks, 37 °C shaking incubator, and centrifuge with rotor and tubes or bottles for harvesting cells)

- Refrigerated centrifuge capable of reaching $\geq 4,500 \times g$ with rotor for the appropriate centrifuge tubes or bottles (not necessary for NucleoBond® Xtra Midi/Maxi Plus kits)
- Centrifugation tubes or vessels with suitable capacity for the volumes specified in the respective protocol
- NucleoBond® Xtra Combi Rack (see ordering information) or equivalent holder

2 Kit specifications

- **NucleoBond® Xtra** kits are suitable for ultra fast purification of plasmids, cosmids, and very large constructs (P1 constructs, BACs, PACs) ranging from 3 kbp up to 300 kbp. For preparation of working solutions and storage conditions see section 5.
- **NucleoBond® Xtra Columns** are polypropylene columns containing **NucleoBond® Xtra Silica Resin** packed between two inert filter elements. The columns are available in Midi and Maxi sizes with typical DNA yields of 500 µg and 1000 µg, respectively.
- All **NucleoBond® Xtra Columns** are resistant to organic solvents such as alcohol, chloroform, and phenol and are also suitable for buffers containing denaturing agents like formamide, urea, or common detergents like Triton X-100 or NP-40.
- **NucleoBond® Xtra Silica Resin** can be used over a wide pH range (pH 2.5–8.5), and can remain in contact with buffers for several hours without any change in its chromatographic properties.
- The **NucleoBond® Xtra Column Filters** are specially designed depth filters that fit into the **NucleoBond® Xtra Columns**. The filters are inserted ready-to-use in the **NucleoBond® Xtra Columns** and allow a time-saving simultaneous clearing of bacterial lysate and loading of cleared lysate onto the **NucleoBond® Xtra Column**. Furthermore, the use of the column filters avoids the time-consuming centrifugation step for lysate clearing.
- The **NucleoBond® Xtra Column Filters** allow complete removal of precipitate even with large lysate volumes without clogging and avoid shearing of large DNA constructs, such as PACs or BACs by the gentle depth filter effect.
- The **NucleoBond® Xtra Midi Plus** and **NucleoBond® Xtra Maxi Plus** kits additionally contain the **NucleoBond® Finalizers** and **NucleoBond® Finalizers Large**, respectively. These tools for a fast concentration and desalination of eluates are suitable for most plasmids and cosmids ranging from 2–50 kbp with recovery efficiencies from 40–90 % (depending on elution volume).
- **NucleoBond® Finalizer** is a polypropylene syringe filter containing a special silica membrane. The **NucleoBond® Finalizer** provides a binding capacity of 500 µg, whereas the **NucleoBond® Finalizer Large** can hold up to 2000 µg plasmid DNA.
- Due to the small dead volumes of the **NucleoBond® Finalizers** the plasmid DNA can be eluted with a concentration up to 3 µg/µL (see section 4.13, Table 4 and Table 5 for dependence of concentration on elution volume).
- All **NucleoBond® Finalizers** are resistant to organic solvents such as alcohol, chloroform, and phenol and are free of endotoxins.

3 About this user manual

The following section 4 provides you with a detailed description of the **NucleoBond® Xtra** purification system and important information about cell growth, cell lysis, and the subsequent purification steps. Sections 5 and 6 inform you about storage, buffer preparation, and safety instructions.

First-time users are strongly advised to read these chapters thoroughly before using this kit. Experienced users can directly proceed with the purification protocols (section 7) or just use the Protocol-at-a-glance for a quick reference.

Section 7 includes the protocols for high copy and low copy plasmid purification as well as for the concentration of **NucleoBond® Xtra** eluates with the **NucleoBond® Finalizer**. This part of the protocol is also available at www.mn-net.com in French and German.

Each procedural step in the purification protocol is arranged like the following example taken from section 7.1:

	Midi	Maxi
5	Cell lysis (Buffer LYS)	
!	Check Lysis Buffer LYS for precipitated SDS prior to use. If a white precipitate is visible, warm the buffer for several minutes at 30–40 °C until precipitate is dissolved completely. Cool buffer down to room temperature (18–25 °C).	
	Add Lysis Buffer LYS to the suspension.	
	Mix gently by inverting the tube 5 times. Do not vortex as this will shear and release contaminating chromosomal DNA from cellular debris into the suspension.	
	Incubate the mixture at room temperature (18–25 °C) for 5 min.	
	<u>Warning:</u> Prolonged exposure to alkaline conditions can irreversibly denature and degrade plasmid DNA and liberate contaminating chromosomal DNA into the lysate.	
	<u>Note:</u> Increase LYS buffer volume proportionally if more than the recommended cell mass is used (see section 4.8 for information on optimal cell lysis).	
	8 mL	12 mL

If you are performing a Midi prep to purify plasmid DNA you will find volumes or incubation times in the white boxes. For Maxi preps please refer to the black boxes.

The name of the buffer, incubation times, repeats or important handling steps are emphasized in **bold type** within the instruction. Additional notes or optional steps are printed in italic. The exclamation point marks information and hints that are essential for a successful preparation.

In the example shown above you are asked to check the Lysis Buffer LYS prior to use and then to lyse the resuspended cell pellet in **8 mL** of **Buffer LYS** when performing a Midi prep and in **12 mL** for a Maxi prep. Follow the handling instructions exactly and note the given hints for protocol alterations.

4 NucleoBond® Xtra plasmid purification system

4.1 Basic principle

The bacterial cells are lysed by an optimized set of newly formulated buffers based on the NaOH/SDS lysis method of Birnboim and Doly*.

After equilibration of the **NucleoBond® Xtra Column** together with the corresponding **NucleoBond® Xtra Column Filter**, the entire lysate is loaded by gravity flow and simultaneously cleared by the specially designed column filter.

Plasmid DNA is bound to the **NucleoBond® Xtra Silica Resin**.

After an efficient washing step the plasmid DNA is eluted, precipitated, and easily dissolved in any suitable buffer (e.g., low-salt buffer or water) for further use.



NucleoBond® Xtra Columns can not be processed by vacuum. Using NucleoBond® Xtra Columns on a vacuum manifold might lead to a complete loss of plasmid DNA.

4.2 NucleoBond® Xtra anion exchange columns

NucleoBond® Xtra is a patented silica-based anion exchange resin, developed by MACHEREY-NAGEL. It is developed for routine separation of different classes of nucleic acids like oligonucleotides, RNA, and plasmids.

NucleoBond® Xtra Silica Resin consists of hydrophilic, macroporous silica beads functionalized with MAE (methyl-amino-ethanol). The dense coating of this functional group provides a high overall positive charge density under acidic pH conditions that permits the negatively charged phosphate backbone of plasmid DNA to bind with high specificity (Figure 1).

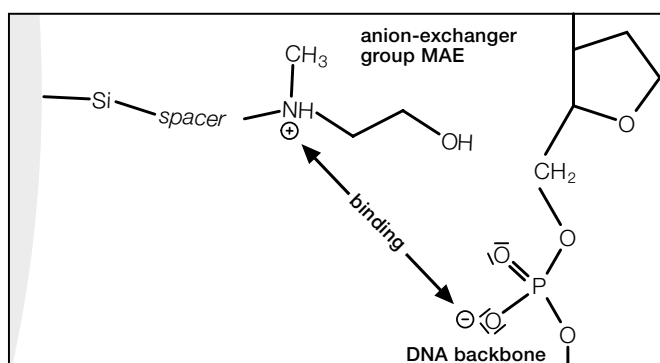


Figure 1 Ionic interaction of the positively charged methyl-hydroxyethyl-amino group with the negative phosphate oxygen of the DNA backbone.

In contrast to the widely used DEAE (diethylaminoethyl) group, the hydroxy group of methyl-hydroxyethyl-amin can be involved in additional hydrogen bonding interactions with the DNA.

* Birnboim, H. C. and Doly, J., (1979) Nucl. Acids Res. 7, 1513–1523

All contaminants from proteins to RNA are washed from the column, the positive charge of the resin is neutralized by a pH shift to slightly alkaline conditions, and pure plasmid DNA is eluted in a high-salt elution buffer.

The purified nucleic acid products are suitable for use in the most demanding molecular biology applications, including transfection, *in vitro* transcription, automated or manual sequencing, cloning, hybridization, and PCR.

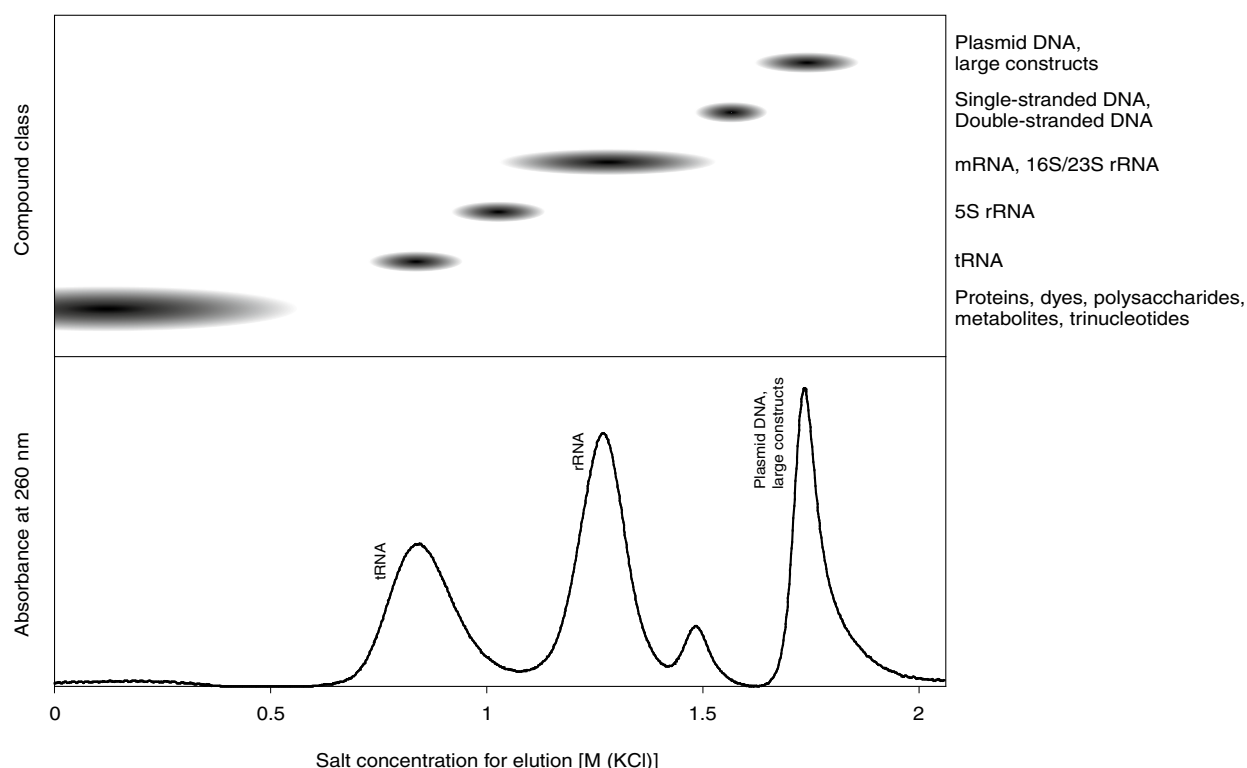


Figure 2 Elution profile of NucleoBond® Xtra Silica Resin at pH 7.0

The more interactions a nucleic acid can form between the phosphate backbone and the positively charged resin the later it is eluted with increasing salt concentration. Large nucleic acids carry more charges than short ones, double stranded DNA more than single stranded RNA.

4.3 Growth of bacterial cultures

Yield and quality of plasmid DNA highly depend on the **type of culture media** and antibiotics, the bacterial host strain, the plasmid type, size, and copy number, but also on the **growth conditions**.

For standard high copy plasmids LB (Luria-Bertani) medium is recommended. The cell culture should be incubated at 37 °C with constant shaking (200–250 rpm) preferably 12–16 h over night. Use flasks of at least three or four times the volume of the culture volume to provide a growth medium saturated with oxygen. Alternatively, rich media like 2xYT (Yeast/Tryptone), TB (Terrific Broth) or CircleGrow can be used. In this case bacteria grow faster, reach the stationary phase much earlier than in LB medium (≤ 2 h), and higher cell masses can be reached. However, this does not necessarily yield more plasmid DNA. Overgrowing a culture might lead to a higher percentage of dead or starving cells and the resulting plasmid DNA might be partially degraded or contaminated with chromosomal DNA. To find the optimal culture conditions, the culture medium and incubation times have to be optimized for each host strain/plasmid construct combination individually.

Cell cultures should be grown under **antibiotic selection** at all times to ensure plasmid propagation. Without this selective pressure, cells tend to lose a plasmid during cell division. Since bacteria grow much faster without the burden of a high copy plasmid, they take over the culture rapidly and the plasmid yield goes down regardless of the cell mass. Table 1 gives information on concentrations of commonly used antibiotics.

Table 1: Information about antibiotics according to Maniatis*

Antibiotic	Stock solution (concentration)	Storage	Working concentration
Ampicillin	50 mg/mL in H ₂ O	-20 °C	20–60 µg/mL
Chloramphenicol	34 mg/mL in EtOH	-20 °C	25–170 µg/mL
Kanamycin	10 mg/mL in H ₂ O	-20 °C	10–50 µg/mL
Streptomycin	10 mg/mL in H ₂ O	-20 °C	10–50 µg/mL
Tetracycline	5 mg/mL in EtOH	-20 °C	10–50 µg/mL
Carbenicillin	50 mg/mL in H ₂ O	-20 °C	20–60 µg/mL

The ***E. coli* host strain** mostly influences the quality of the plasmid DNA. Whereas strains like DH5a® or XL1-Blue usually produce high quality super-coiled plasmid DNA, other strains like for example HB101 with high levels of endonuclease activity might yield lower quality plasmid giving poor results in downstream applications like enzymatic restriction or sequencing.

The **type of plasmid**, especially the **size and the origin of replication (ori)** has a crucial influence on DNA yield. In general, the larger the plasmid or the cloned insert is, the lower is the expected DNA yield due to a lower copy number. Even a high copy construct based on a ColE1 ori can behave like a low copy vector in case of a large or unfavorable insert. In addition, the ori itself influences the yield by factor 10–100. Thus plasmids based on for example pBR322 or pACYC, cosmids or BACs are maintained at copy numbers < 20 down to even only 1, whereas vectors based on for example pUC, pBluescript or pGEM can be present in several hundred copies per cell.

Therefore, all the above mentioned factors should be taken into consideration if a particular DNA yield has to be achieved. Culture volume and lysis procedures have to be adjusted accordingly.

* Maniatis T, Fritsch EF, Sambrook J: Molecular cloning. A laboratory manual, Cold Spring Harbor, Cold Spring, New York 1982.

4.4 Chloramphenicol amplification of low copy plasmids

To dramatically increase the low copy number of pMB1/colE1 derived plasmids grow the cell culture to mid or late log phase ($OD_{600} \approx 0.6\text{--}2.0$) under selective conditions with an appropriate antibiotic. Then add 170 $\mu\text{g/mL}$ chloramphenicol and continue incubation for a further 8–12 hours. Chloramphenicol inhibits host protein synthesis and thus prevents replication of the host chromosome. Plasmid replication, however, is independent of newly synthesized proteins and continues for several hours until up to 2000–3000 copies per cell are accumulated*.

Alternatively, the cell culture can be grown with only partial inhibition of protein synthesis under low chloramphenicol concentrations (10–20 $\mu\text{g/mL}$) resulting in a 5–10-fold greater yield of plasmid DNA**.

Both methods show the positive side effect of much less genomic DNA per plasmid, but they obviously work only with plasmids that do not carry the chloramphenicol resistance gene. Furthermore, the method is only effective with low copy number plasmids under stringent control (e.g., pBR322). All modern high copy number plasmids (e.g., pUC) are already under relaxed control due to mutations in the plasmid copy number control genes and show no significant additional increase in their copy number.

4.5 Culture volume for high copy plasmids

Due to the influence of growth media (TB, CircleGrow, 2 xYT), growth conditions (shaking, temperature), host strain, or type of plasmid insert etc. the final amount of cells in a bacterial culture can vary over a wide range. By rule of thumb, 1 liter of *E. coli* LB culture with an OD_{600} of 1 consists of 1×10^{12} cells and yields about 1.5–1.8 g cell wet weight. Overnight cultures grown in LB medium usually reach an OD_{600} of 3–6 under vigorous shaking in flasks. Fermentation cultures even reach an OD_{600} of 10 and more. The expected DNA yield for a high copy plasmid is approximately 1 mg per gram cell wet weight.

It is therefore important to **adjust the cell mass rather than the culture volume** for the best plasmid purification results. But since the cell mass or cell wet weight is tedious to determine it was replaced in this manual by the mathematical product of optical density at 600 nm (OD_{600}) and culture volume (Vol) - two variables that are much easier to measure.

$$ODV = OD_{600} \times Vol \text{ [mL]}$$

Note that for a correct OD determination the culture samples have to be diluted if OD_{600} exceeds 0.5 in order to increase proportionally with cell mass. For a well grown *E. coli* culture a 1:10 dilution with fresh culture medium is recommended. The measured OD_{600} is then multiplied with the dilution factor 10 to result in a theoretical OD_{600} value. This OD_{600} is used in Table 2 to determine the appropriate culture volume. Table 2 shows recommended ODVs and the corresponding pairs of OD_{600} and culture volume that can be easily handled using the standard kit protocol lysis buffer volumes. For example, if the OD_{600} of your *E. coli* culture is 6, use 66 mL culture for a Midi prep or 200 mL for a Maxi prep.

* Maniatis T, Fritsch EF, Sambrook J: Molecular cloning. A laboratory manual, Cold Spring Harbor, Cold Spring, New York 1982.

** Frenkel L, Bremer H: Increased amplification of plasmids pBR322 and pBR327 by low concentrations of chloramphenicol, DNA (5), 539 – 544, 1986..

Table 2: Recommended culture volumes for high copy plasmids							
NucleoBond® Xtra	Pellet wet weight	Rec. ODV	OD ₆₀₀ = 2	OD ₆₀₀ = 4	OD ₆₀₀ = 6	OD ₆₀₀ = 8	OD ₆₀₀ = 10
Midi	0.75 g	400	200 mL	100 mL	66 mL	50 mL	40 mL
Maxi	2.25 g	1200	600 mL	300 mL	200 mL	150 mL	120 mL

4.6 Culture volume for low copy plasmids

NucleoBond® Xtra kits are designed for isolation of high copy plasmids (up to several hundred copies/cell) as well as low copy plasmids (< 20 copies/cell). However, when purifying low copy plasmids, the cell mass and the lysis buffer volumes should be increased at least by factor 2 to provide enough DNA to utilize the columns' binding capacity. Table 3 shows recommended ODVs and the corresponding pairs of OD₆₀₀ and culture volume for low copy plasmid cell cultures (for detailed information on calculating ODV = OD₆₀₀ x Vol. refer to section 4.5). For example, if the OD₆₀₀ of your *E. coli* culture is 6, use 133 mL culture for a Midi prep or 400 mL for a Maxi prep.

Table 3: Recommended culture volumes for low copy plasmids							
NucleoBond® Xtra	Pellet wet weight	Rec. ODV	OD ₆₀₀ = 2	OD ₆₀₀ = 4	OD ₆₀₀ = 6	OD ₆₀₀ = 8	OD ₆₀₀ = 10
Midi	1.5 g	800	400 mL	200 mL	133 mL	100 mL	80 mL
Maxi	4.5 g	2400	1200 mL	600 mL	400 mL	300 mL	240 mL

For higher yields, it is advantageous to increase the cell culture and lysis buffer volumes even more (e.g., by factor 3–5). In this case additional lysis buffer can be ordered separately (see ordering information). Furthermore, a centrifuge should be used for lysate clarification instead of the provided **NucleoBond® Xtra Column Filters** since their capacity for precipitate is limited.

Alternatively, chloramphenicol amplification can be considered to increase the plasmid copy number (see section 4.4)

4.7 Lysate neutralization and LyseControl

Proper mixing of the lysate with Neutralization Buffer NEU is of utmost importance for complete precipitation of SDS, protein, and genomic DNA. Incomplete neutralization leads to reduced yields, slower flow-rates, and potential clogging of the **NucleoBond® Xtra** Column Filter. However, released plasmid DNA is very vulnerable at this point and shaking too much or too strongly will damage the DNA.

Therefore, **do not vortex or shake** but **just invert the mixture very gently** until a fluffy off-white precipitate has formed and the **LyseControl** has turned **colorless** throughout the lysate without any traces of blue color.

4.8 Cell lysis

The bacterial cell pellet is resuspended in Buffer RES and lysed by a sodium hydroxide/SDS treatment with Buffer LYS. Proteins, as well as chromosomal and plasmid DNA are denatured under these conditions. RNA is degraded by RNase A. Neutralization Buffer NEU, containing potassium acetate, is then added to the lysate, causing SDS to precipitate as KDS (potassium dodecyl sulfate) and pulling down proteins, chromosomal DNA, and other cellular debris. The potassium acetate buffer also neutralizes the lysate. Plasmid DNA can revert to its native super-coiled structure and remains in solution.

The **NucleoBond® Xtra** buffer volumes (standard protocol) are adjusted to ensure optimal lysis for culture volumes, appropriate for high copy plasmids according to section 4.5, Table 2. Using too much cell material leads to inefficient cell lysis and precipitation and might reduce plasmid yield and purity. Therefore, lysis buffer volumes should be increased when applying larger culture volumes in case of for example low copy vector purification (section 4.6, Table 3).

By rule of thumb, calculate the necessary lysis buffer volumes for RES, LYS, and NEU as follows:

$$\text{Vol. [mL]} = \text{Culture Volume [mL]} \times \text{OD}_{600} / 50$$

For example, if 200 mL of a low copy bacterial culture ($\text{OD}_{600} = 4$) is to be lysed, the appropriate volumes of lysis buffers RES, LYS, and NEU are 16 mL each. If more lysis buffer is needed than is provided with the kit, an additional buffer set including buffers RES, LYS, NEU, and RNase A can be ordered separately (see ordering information).

By using sufficient amounts of lysis buffer, lysis time can be limited to 3–4 minutes and should not exceed 5 minutes. Prolonged exposure to alkaline conditions can irreversibly denature and degrade plasmid DNA and liberate contaminating chromosomal DNA into the lysate.

Please note that the calculated lysis buffer volumes for NucleoBond® Xtra Maxi preparations do not match the recommended volumes in the protocol due to the fact that most users start with much less cells than the recommended ODV 1200. Furthermore, the 3 x 12 mL of the protocol can conveniently be used in combination with 50 mL centrifugation tubes. More lysis buffer usually requires to split the sample.

4.9 Difficult-to-lyse strains

For plasmid purification of for example Gram-positive bacteria or strains with a more resistant cell wall it might be advantageous to start the preparation with a lysozyme treatment. Therefore, resuspend the cell pellet in Buffer RES containing **2 mg/mL lysozyme** and incubate at **37 °C** for **30 minutes**. Proceed then with the lysis procedure according to the **NucleoBond® Xtra** standard protocol.

4.10 Setup of NucleoBond® Xtra Columns

Ideally the **NucleoBond® Xtra Midi** or **Maxi Columns** are placed into a **NucleoBond® Xtra Combi Rack** (see ordering information). They are held either by the collar ring of the cartridges or by the Plastic Washers (reusable) included in the kit to individually adjust the height of each column (see Figure 3). The Plastic Washers (reusable) can also be used to hold the columns on top of suitable collection tubes or flasks. The **NucleoBond® Xtra Combi Rack** can be used in combination with **NucleoBond® PC 100, 500, and 2000** as well. Note that the **NucleoBond® Xtra Midi Columns** can also be placed in the **NucleoBond® Rack Large** (REF 740563).

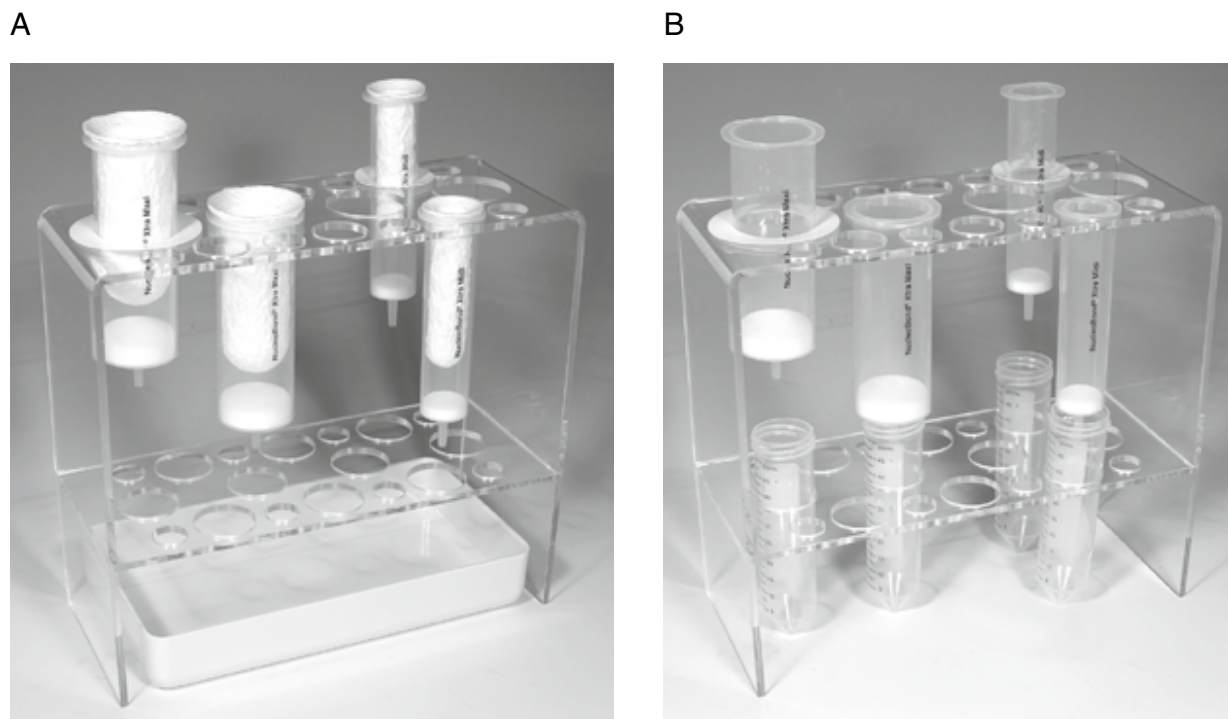
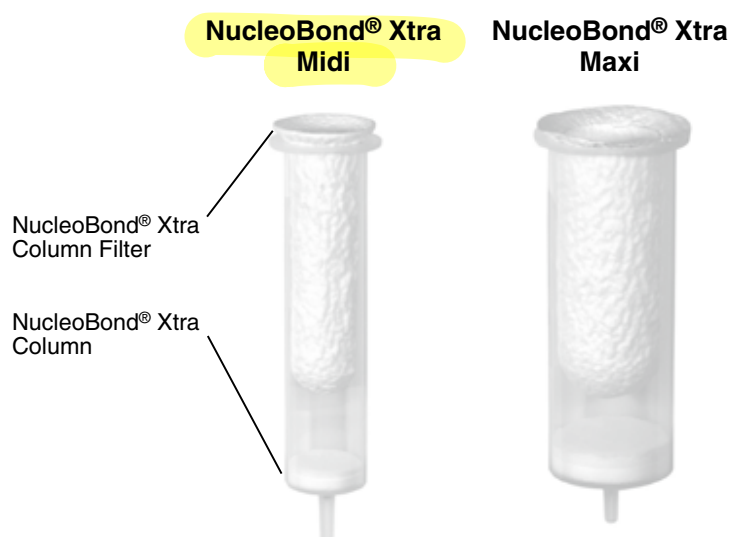


Figure 3 Setup of NucleoBond® Xtra Midi/Maxi Columns with the NucleoBond® Xtra Combi Rack

A: Setup for clarification, loading, and first washing step; B: Setup for elution.

4.11 Filtration and loading of the lysate

After the alkaline lysis, the sample has to be cleared from cell debris and precipitate to ensure high plasmid purity and a fast column flow rate. This is achieved by passing the solution through a **NucleoBond® Xtra Column Filter** which is provided already inserted into the **NucleoBond® Xtra Column**.



The **NucleoBond® Xtra Column Filters** are designed to eliminate the centrifugation step after alkaline lysis. They are pre-wet during column equilibration and allow a time-saving simultaneous clearing of bacterial lysate and loading of the **NucleoBond® Xtra Column**.

Compared to lysate clearing by centrifugation or syringe filters the **NucleoBond® Xtra Column Filter** furthermore avoids shearing of large DNA constructs such as PACs or BACs by the gentle depth filter effect (filtration occurs on the surface of the filter as well as inside the filter matrix). Its special material and design lead to very rapid passage of the lysate through the filter and even very large lysate volumes can be applied without the risk of clogging. This is especially important for low copy plasmid purification for example. However, if more than the recommended cell mass (see section 4.5, Table 2, section 4.6, Table 3) was lysed, it might be advantageous to use a centrifuge for lysate clarification rather than the provided column filters due to their limited precipitate capacity.

4.12 Washing of the column

The high salt concentration of the lysate prevents proteins and RNA from binding to the **NucleoBond® Xtra Column** (see section 4.2, Figure 2). However, to remove all traces of contaminants and to purge the dead volume of the **NucleoBond® Xtra Column Filters** it is important to wash the column and the filter in two subsequent washing steps.

First apply **Equilibration Buffer EQU** to the funnel rim of the filter to wash all residual lysate out of the filter onto the column. Do not just pour the buffer inside the filter. Then pull out and discard the column filter or remove the filter by turning the column upside down. It is essential to wash the **NucleoBond® Xtra Column** without filter for a second time with **Wash Buffer WASH**. This ensures highest yields with best achievable purity.

4.13 Elution and concentration of plasmid DNA

Elution is carried out under high-salt conditions and by a shift of pH from 7.0 to 9.0. Under these alkaline conditions the positive charge of the anion exchange resin is neutralized and plasmid DNA is released. For any downstream application it is necessary to precipitate the DNA and to remove salt and all traces of alcohol since they disturb or inhibit enzymatic activity needed for restriction or sequencing reactions.

All **NucleoBond® Xtra** eluates already contain enough salt for an isopropanol precipitation of DNA. Therefore the precipitation is started by directly adding 0.7 volumes of isopropanol. To prevent co-precipitation of salt, use **room-temperature (18–25°C) isopropanol** only and do not let the plasmid DNA solution drop into a vial with isopropanol but **add isopropanol to the final eluate and mix immediately**.

Afterwards either follow the centrifugation protocol given after the **NucleoBond® Xtra** purification protocol or follow the support protocol for the **NucleoBond® Finalizers** in section 7.3 to eliminate the time-consuming centrifugation steps for precipitation (use of **NucleoBond® Finalizers** is only recommended for vector sizes smaller than 50 kbp).

The **NucleoBond® Finalizers** are designed for quick concentration and desalination of plasmid and cosmid DNA eluates that are obtained by anion exchange chromatography based DNA purification. The sample is precipitated with isopropanol as mentioned above and loaded onto a special silica membrane by means of a syringe. After an ethanolic washing step the membrane is dried by pressing air through the filter. Elution of pure DNA is carried out with slightly alkaline low salt buffers like **Buffer TRIS** (5 mM Tris/HCl, pH 8.5, provided with all **NucleoBond® Xtra Plus** kits) or TE buffer (10 mM Tris/HCl, pH 7.5, 1 mM EDTA). Do not use pure water unless pH is definitely higher than 7.0

For maximum yield it is recommended to perform the elution step twice. The first elution step is done using fresh buffer whereas in the second elution step the eluate from

the first elution is reapplied on the **NucleoBond® Finalizer** to allow complete solubilization of the plasmid.

DNA recovery highly depends on the used elution buffer volume. Large volumes result in a high recovery of up to 90 % but in a lower DNA concentration. Small elution volumes on the other hand increase the concentration but at the cost of DNA yield.

If a small volume is chosen, make sure to recover as much eluate as possible from the syringe and **NucleoBond® Finalizer** by pressing air through the **NucleoBond® Finalizer** several times after elution and collecting every single droplet to minimize the dead volume.

Figure 4 and Figure 5 illustrate exemplarily how DNA recovery and final DNA concentration depend on the buffer volume which is used for elution of DNA from the **NucleoBond® Finalizer** and **NucleoBond® Finalizer Large**, respectively.

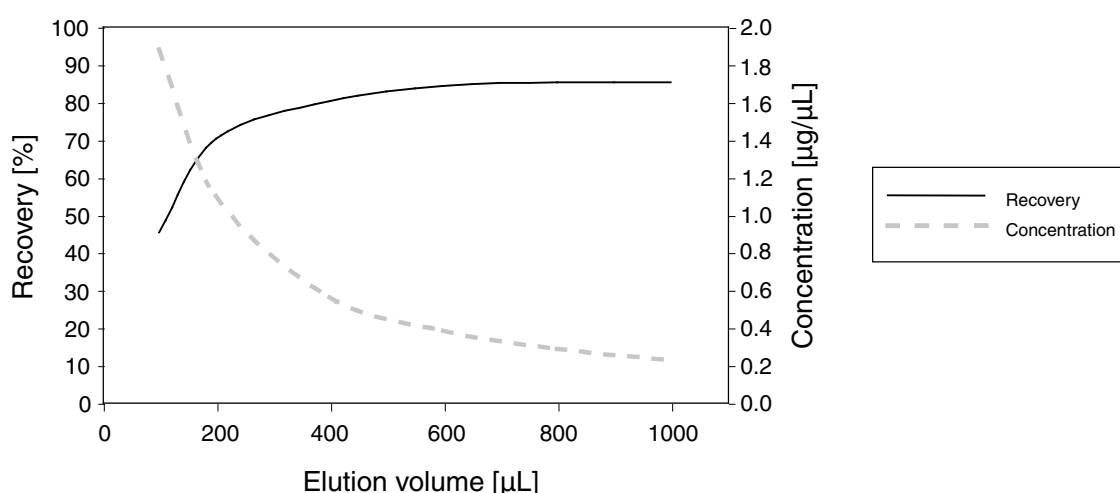


Figure 4 Final DNA recovery and concentration after NucleoBond® Finalizer application

A NucleoBond® Xtra Midi eluate containing 250 µg plasmid DNA (8 kbp) was loaded onto a NucleoBond® Finalizer and eluted two-fold with increasing volumes of TE buffer.

The **NucleoBond® Finalizer** is designed to hold a maximum of 500 µg DNA and is therefore ideally suited to be used in combination with **NucleoBond® Xtra Midi**. Maximum DNA recovery can be achieved by using > 600 µL of elution buffer. For a higher concentration experienced users can lower the elution buffer volume to 400–200 µL.

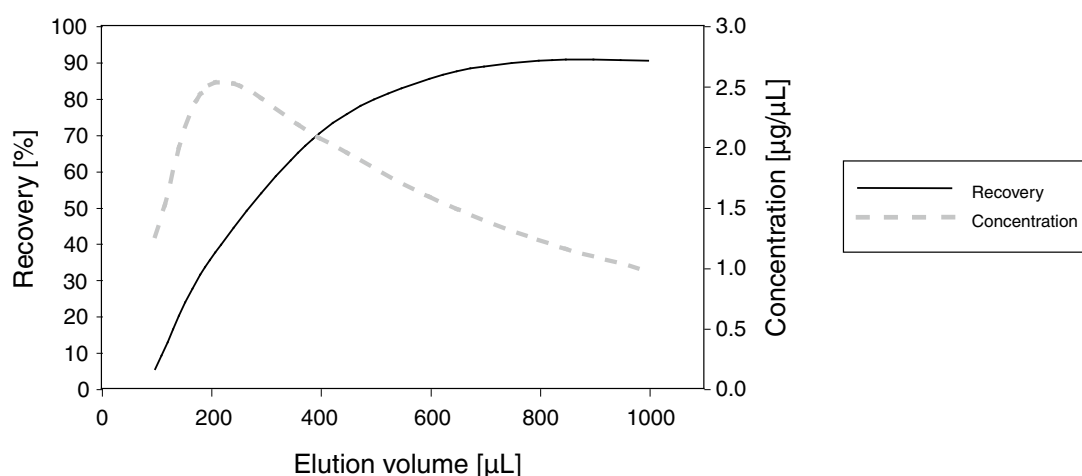
Table 4 gives an overview about recovery and concentration of different amounts of plasmid DNA loaded onto a **NucleoBond® Finalizer**. DNA was eluted two-fold with increasing volumes of TE. Please refer to this tables to select an elution buffer volume that meets your needs best.

Table 4: DNA recovery and concentration for the NucleoBond® Finalizer

		Elution volume					
		100 µL	200 µL	400 µL	600 µL	800 µL	1000 µL
Loaded DNA	500 µg	35 %	60 %	70 %	75 %	75 %	75 %
		2.5 µg/µL	2.3 µg/µL	1.2 µg/µL	0.8 µg/µL	0.6 µg/µL	0.5 µg/µL
	250 µg	40 %	65 %	75 %	80 %	80 %	80 %
		1.9 µg/µL	1.1 µg/µL	0.6 µg/µL	0.4 µg/µL	0.3 µg/µL	0.2 µg/µL
	100 µg	45 %	70 %	80 %	85 %	85 %	85 %
		0.7 µg/µL	0.4 µg/µL	0.2 µg/µL	0.1 µg/µL	0.1 µg/µL	0.1 µg/µL
	50 µg	30 %	75 %	85 %	90 %	90 %	90 %
		0.3 µg/µL	0.2 µg/µL	0.1 µg/µL	0.1 µg/µL	0.1 µg/µL	< 0.1 µg/µL

DNA recovery

DNA concentration

**Figure 5 Final DNA recovery and concentration after NucleoBond® Finalizer Large application**

A NucleoBond® Xtra Maxi eluate containing 1000 µg plasmid DNA (8 kbp) was loaded onto a NucleoBond® Finalizer Large and eluted two-fold with increasing volumes of TE buffer.

NucleoBond® Xtra Maxi eluates are easily concentrated with a **NucleoBond® Finalizer Large** which is able to bind up to 2000 µg plasmid DNA. Maximum DNA recovery can be achieved by using > 800 µL of elution buffer. For a higher concentration experienced users can lower the elution buffer volume to 600–400 µL.

Table 5 gives an overview about recovery and concentration of different amounts of plasmid DNA loaded onto a **NucleoBond® Finalizer Large**. DNA was eluted two-fold with increasing volumes of TE. Please refer to this tables to select an elution buffer volume that meets your needs best.

Table 5: DNA recovery and concentration for the NucleoBond® Finalizer Large

		Elution volume					
		100 µL	200 µL	400 µL	600 µL	800 µL	1000 µL
Loaded DNA	1500 µg	5 %	30 %	65 %	80 %	85 %	90 %
		1.9 µg/µL	3.2 µg/µL	2.9 µg/µL	2.2 µg/µL	1.7 µg/µL	1.4 µg/µL
	1000 µg	5 %	35 %	70 %	85 %	90 %	90 %
		1.3 µg/µL	2.5 µg/µL	2.1 µg/µL	1.6 µg/µL	1.2 µg/µL	1.0 µg/µL
	500 µg	10 %	40 %	70 %	85 %	90 %	90 %
		1.3 µg/µL	1.4 µg/µL	1.0 µg/µL	0.8 µg/µL	0.6 µg/µL	0.5 µg/µL
	100 µg	15 %	45 %	70 %	80 %	85 %	90 %
		0.4 µg/µL	0.3 µg/µL	0.2 µg/µL	0.1 µg/µL	0.1 µg/µL	0.1 µg/µL

DNA recovery

DNA concentration

4.14 Determination of DNA yield and quality

The **yield** of a plasmid preparation should be estimated prior to and after the isopropanol precipitation in order to calculate the recovery after precipitation and to find the best volume to dissolve the pellet in. Simply use either **NucleoBond® Xtra** Elution Buffer ELU or the respective low-salt buffer as a blank in your photometric measurement.

The nucleic acid **concentration** of the sample can be calculated from its UV absorbance at 260 nm where an absorbance of 1 (1 cm path length) is equivalent to 50 µg DNA/mL. Note that the absolute measured absorbance should lie between 0.1 and 0.7 in order to be in the linear part of Lambert-Beer's law. Dilute your sample in the respective buffer if necessary.

The plasmid **purity** can be checked by UV spectroscopy as well. A ratio of A_{260}/A_{280} between 1.80–1.90 and A_{260}/A_{230} around 2.0 indicates pure plasmid DNA. An A_{260}/A_{280} ratio above 2.0 is a sign for too much RNA in your preparation, an A_{260}/A_{280} ratio below 1.8 indicates protein contamination.

Plasmid **quality** can be checked by running the precipitated samples on a 1 % agarose gel. This will give information on conformation and structural integrity of isolated plasmid DNA, i. e. it shows whether the sample is predominantly present in the favorable super-coiled form (ccc, usually the fastest band), as an open circle (oc), or even in a linear form (see section 8.1, Figure 6).

4.15 Convenient stopping points

Cell pellets can easily be stored for several months at -20 °C.

Cleared lysates can be kept on ice or at 4 °C for several days.

For optimal performance the column purification should not be interrupted. However, the columns can be left unattended for several hours since the columns do not run dry. This might cause only small losses in DNA yield.

The eluate can be stored for several days at 4 °C. Note that the eluate should be warmed up to room temperature before precipitating the DNA to avoid co-precipitation of salt.

5 Storage conditions and preparation of working solutions

All kit components can be stored at room temperature (18–25 °C) and are stable for at least two years.

Storage of Buffer LYS below 20 °C may cause precipitation of SDS. If salt precipitate is observed, incubate buffer at 30–40 °C for several minutes and mix well until all precipitate is redissolved completely. Cool down to room temperature before use.

Before the first use of the **NucleoBond® Xtra Midi / Maxi** kit, prepare the following:

- Dissolve the lyophilized RNase A* by the addition of 1 mL of Buffer RES. Wearing gloves is recommended. Pipette up and down until the RNase A is dissolved completely. Transfer the RNase A solution back to the bottle containing Buffer RES and shake well. Note the date of RNase A addition. The final concentration of RNase A is 60 µg/mL Buffer RES. Store Buffer RES with RNase A at 4 °C. The solution will be stable at this temperature for at least 6 months.

* REF 740414.100 contains 2 x 50 mg of RNase A. Make sure to dissolve RNase A of both vials, each in 1 mL of Buffer RES, and transfer the solution back into the bottle containing Buffer RES.

6 Safety instructions

The following components of the **NucleoBond® Xtra** kit contain hazardous contents.

Wear gloves and goggles and follow the safety instructions given in this section.

GHS classification

Only harmful features need not be labeled with H and P phrases up to 125 mL or 125 g.
Mindergefährliche Eigenschaften müssen bis 125 mL oder 125 g nicht mit H- und P-Sätzen gekennzeichnet werden.

Component	Hazard contents	GHS symbol	Hazard phrases	Precaution phrases
Inhalt	Gefahrstoff	GHS Symbol	H-Sätze	P-Sätze
LYS	Sodium hydroxide solution 0.5–1.0 % <i>Natriumhydroxid-Lösung</i> 0,5–1,0 %	 WARNING ACHTUNG	315, 319	280sh
WASH	Ethanol 5–20 % <i>Ethanol 5–20 %</i>	 WARNING ACHTUNG	226	210
ELU	2-propanol 10–15 % <i>2-Propanol 10–15 %</i>	 WARNING ACHTUNG	226, 319	210, 280sh
RNase A	RNase A 90–100 % <i>RNase A 90–100 %</i>	 DANGER GEFAHR	334	261sh, 342+311



The symbol shown on labels refers to further safety information in this section.
Das auf Etiketten dargestellte Symbol weist auf weitere Sicherheitsinformationen dieses Kapitels hin.

Hazard phrases

H 226	Flammable liquid and vapour. <i>Flüssigkeit und Dampf entzündbar.</i>
H 315	Causes skin irritation. <i>Verursacht Hautreizungen.</i>
H 319	Causes serious eye irritation. <i>Verursacht schwere Augenreizung.</i>
H 334	May cause allergy or asthma symptoms or breathing difficulties if inhaled. <i>Kann bei Einatmen Allergie, asthmaartige Symptome oder Atembeschwerden verursachen.</i>

Precaution phrases

- P 210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
Von Hitze, heißen Oberflächen, Funken, offenen Flammen sowie anderen Zündquellenarten fernhalten. Nicht rauchen.
- P 261sh Avoid breathing dust/vapors.
Einatmen von Staub/Dampf vermeiden.
- P 280sh Wear protective gloves / eye protection.
Schutzhandschuhe / Augenschutz tragen.
- P 342+311 If experiencing respiratory symptoms: Call a POISON CENTER / doctor.
Bei Symptomen der Atemwege: GIFTINFORMATIONSZENTRUM / Arzt anrufen.

For further information please see Material Safety Data Sheets (www.mn-net.com).
Weiterführende Informationen finden Sie in den Sicherheitsdatenblättern (www.mn-net.com).

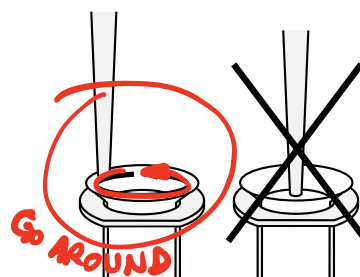
7 NucleoBond® Xtra plasmid purification

The following section includes the protocols for high copy and low copy plasmid purification as well as for concentration of NucleoBond® Xtra eluates with the NucleoBond® Finalizers.

7.1 High copy plasmid purification (Midi, Maxi)

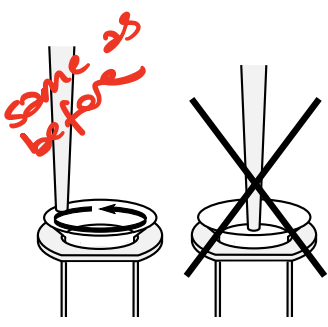
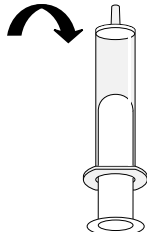
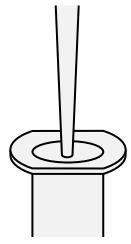
	Midi	Maxi
1 Preparation of starter culture <i>only if starting from plates</i>	Inoculate a 3–5 mL starter culture of LB medium with a single colony picked from a freshly streaked agar plate. Make sure that plate and liquid culture contain the appropriate selective antibiotic to guarantee plasmid propagation (see section 4.3 for more information). Shake at 37 °C and ~ 300 rpm for ~ 8 h.	
2 Preparation of large overnight culture <i>start from here with glycerol stock</i>	Inoculate an overnight culture by diluting the starter culture 1/1000 into the given volumes of LB medium also containing the appropriate selective antibiotic. Grow the culture overnight at 37 °C and ~ 300 rpm for 12–16 h. <i>(1:1000) Carbenicillin 100 mg/ml → 100 µg/ml</i>	
	100 mL <i>+ 10 µg carbenicillin</i>	300 mL
3 Harvest of bacterial cells <i>Just check bacteria have grown [MEDIUM IS CLOUDY]</i>	Measure the cell culture OD ₆₀₀ and determine the recommended culture volume.	
	$V \text{ [mL]} = \frac{400}{OD_{600}}$	$V \text{ [mL]} = \frac{1200}{OD_{600}}$
	Pellet the cells by centrifugation at 4,500–6,000 x g for ≥ 10 min at 4 °C and discard the supernatant completely.	
	<i>Note: Optimal lysis conditions are achieved by a unique ratio of lysis buffer RES, LYS, and NEU to the cell mass. See section 4.5 for recommendations concerning the optimal pelleted culture volume for cells containing high copy plasmids and section 4.6 for recommendations concerning cells with low copy plasmids. Read section 4.5 carefully as excess cell input will result in reduced yield. The recommended culture volumes below are calculated for a final OD₆₀₀ of around 4.</i>	
4 Resuspension (Buffer RES) <i>move to one tube</i>	Resuspend the cell pellet completely in Resuspension Buffer RES + RNase A by pipetting the cells up and down or vortexing the cells.	
	For an efficient cell lysis it is important that no clumps remain in the suspension.	
	<i>Note: Increase RES buffer volume proportionally if more than the recommended cell mass is used (see section 4.8 for information on optimal cell lysis and section 4.9 regarding difficult-to-lyse strains).</i>	
	8 mL TOTAL	12 mL

	Midi	Maxi
5	Cell lysis (Buffer LYS) Check Lysis Buffer LYS for precipitated SDS prior to use. If a white precipitate is visible, warm the buffer for several minutes at 30–40 °C until precipitate is dissolved completely. Cool buffer down to room temperature (18–25 °C). Add Lysis Buffer LYS to the suspension. Mix gently by inverting the tube 5 times. Do not vortex as this will shear and release contaminating chromosomal DNA from cellular debris into the suspension. Incubate the mixture at room temperature (18–25 °C) for 5 min. <i>Warning:</i> Prolonged exposure to alkaline conditions can irreversibly denature and degrade plasmid DNA and liberate contaminating chromosomal DNA into the lysate. <i>Note:</i> Increase LYS buffer volume proportionally if more than the recommended cell mass is used (see section 4.8 for information on optimal cell lysis).	
	8 mL	12 mL
6	Equilibration (Buffer EQU) Equilibrate a NucleoBond® Xtra Column together with the inserted column filter with Equilibration Buffer EQU. Apply the buffer onto the <u>rim</u> of the column filter as shown in the picture and make sure to wet the entire filter. Allow the column to empty by gravity flow. The column does not run dry.	
	12 mL	25 mL



	Midi	Maxi
7 Neutralization (Buffer NEU)		
! Add Neutralization Buffer NEU to the suspension and immediately mix the lysate gently by inverting the tube until blue samples turns colorless completely! Do not vortex. ■ The flask or tube used for this step should not be filled more than two thirds to allow homogeneous mixing. Make sure to neutralize completely to precipitate all the protein and chromosomal DNA. The lysate should turn from a slimy, viscous consistency to a low viscosity, homogeneous suspension of an off-white flocculate. In addition, LyseControl should turn completely colorless without any traces of blue. Immediately proceed with step 8. An incubation of the lysate is not necessary. <i>Note: Increase NEU buffer volume proportionally if more than the recommended cell mass is used (see section 4.8 for information on optimal cell lysis).</i>		
	8 mL	12 mL
8 Clarification and loading		
! Make sure to have a homogeneous suspension of the precipitate by inverting the tube 3 times directly before applying the lysate to the equilibrated NucleoBond® Xtra Column Filter to avoid clogging of the filter. ■ The lysate is simultaneously cleared and loaded onto the column. Refill the filter if more lysate has to be loaded than the filter is able to hold. Allow the column to empty by gravity flow. Alternative: The precipitate can be removed by centrifugation at $\geq 5,000 \times g$ for at least 10 min, for example if more than double the recommended cell mass was used. If the supernatant still contains suspended matter transfer it to a new tube and repeat the centrifugation, preferably at higher speed, or apply the lysate to the equilibrated NucleoBond® Xtra Column Filter. This clarification step is extremely important since residual precipitate may clog the NucleoBond® Xtra Column. To load the column you can either apply the cleared lysate to the equilibrated filter or remove the unused filter beforehand. Allow the column to empty by gravity flow. <i>Note: You may want to save all or part of the flow-through for analysis (see section 8.1).</i>		

SHOULD
BE 24 mL
BY NOW

	Midi	Maxi
9	1st Wash: Column filter and column (Buffer EQU) ! Wash the NucleoBond® Xtra Column Filter and NucleoBond® Xtra Column with Equilibration Buffer EQU. Apply the buffer to the funnel shaped rim of the filter and make sure it is washing out the lysate which is remaining in the filter. Omitting this step or just pouring the buffer directly inside the funnel may reduce plasmid yield. 5 mL	Maxi 
10	Filter removal Either pull out the NucleoBond® Xtra Column Filter or discard it by turning the column upside down.	
11	2nd Wash: Column only (Buffer WASH) ! Wash the NucleoBond® Xtra Column with Wash Buffer WASH. It is important to remove the column filter before applying the washing buffer to avoid low purity. 8 mL	 25 mL
12	Elution (Buffer ELU) <i>change collection tube to a new one</i> Elute the plasmid DNA with Elution Buffer ELU. Collect the eluate in a 15 mL or 50 mL centrifuge tube (not provided). <i>Note: Preheating Buffer ELU to 50 °C prior to elution may improve yields for large constructs such as BACs.</i> <i>Optional: Determine plasmid yield by UV spectrophotometry in order to adjust desired concentration of DNA in step 15 and calculate the recovery after precipitation.</i> 5 mL	15 mL
	! NucleoBond® Xtra Midi/Maxi Plus: Proceed with step 13 for the centrifugation protocol after isopropanol precipitation or continue with section 7.3 for plasmid concentration and desalting by using the NucleoBond® Finalizer (NucleoBond® Xtra Midi Plus) or NucleoBond® Finalizer Large (NucleoBond® Xtra Maxi Plus).	

	Midi	Maxi
13 Precipitation		
	Add room-temperature isopropanol to precipitate the eluted plasmid DNA.	
	Vortex thoroughly!	
	Centrifuge at $\geq 4,500 \times g$ for ≥ 15 min at $\leq$ room temperature, preferably at $\geq 15,000 \times g$ for 30 min at 4 °C. Carefully discard the supernatant.	
	3.5 mL	10.5 mL
14 Washing and drying		
	Add room-temperature 70 % ethanol to the pellet.	
	2 mL	4 mL
	Centrifuge at $\geq 4,500 \times g$, preferably $\geq 15,000 \times g$ for 5 min at room temperature (18–25 °C).	
	Carefully remove ethanol completely from the tube with a pipette tip. Allow the pellet to dry at room temperature (18–25 °C).	
	<i>Note: Plasmid DNA might be harder to dissolve when over-dried.</i>	
	10–15 min	15–30 min
15 Reconstitution		
	Dissolve the DNA pellet in an appropriate volume of buffer TE or sterile H ₂ O. Depending on the type of centrifugation tube, dissolve under gentle pipetting up and down or constant spinning in a sufficient amount of buffer for 10–60 min (3D-shaker).	
	Determine plasmid yield by UV spectrophotometry. Confirm plasmid integrity by agarose gel electrophoresis (see section 4.14).	

- usually dilute first in 50-70 μ l
- measure at nanodrop
- bring to 1000 ng/ μ l
- freeze at -20°C

7.2 Low copy plasmid purification (Midi, Maxi)

The lysis buffer volumes provided in the kit are adjusted for high copy plasmid purification. Therefore, additional buffer has to be ordered separately for routine purification of low copy plasmids (see ordering information).

	Midi	Maxi
1 Preparation of starter culture	Inoculate a 3–5 mL starter culture of LB medium with a single colony picked from a freshly streaked agar plate. Make sure that plate and liquid culture contain the appropriate selective antibiotic to guarantee plasmid propagation (see section 4.3 for more information). Shake at 37 °C and ~ 300 rpm for ~ 8 h.	
2 Preparation of large overnight culture	! <i>Note: To utilize the entire large binding capacity of the NucleoBond® Xtra Columns it is important to provide enough plasmid DNA. For the standard low copy procedure the culture volumes were doubled compared to the high copy vector protocol. However, due to a plasmid content that is 10–100 times lower, this might be insufficient. If you need large amounts of low copy plasmids, further increase the culture volume by factor 3–5. The recommended culture volumes below are calculated for a final OD₆₀₀ of around 4 (see section 4.6 for more information).</i> Inoculate an overnight culture by diluting the starter culture 1/1000 into the given volumes of LB medium also containing the appropriate selective antibiotic. Grow the culture overnight at 37 °C and ~ 300 rpm for 12–16 h.	
	200 mL	600 mL
3 Harvest of bacterial cells	Measure the cell culture OD ₆₀₀ and determine the recommended culture volume.	
	$V \text{ [mL]} = \frac{800}{OD_{600}}$	$V \text{ [mL]} = \frac{2400}{OD_{600}}$
	Pellet the cells by centrifugation at 4,500–6,000 x g for ≥ 10 min at 4 °C and discard the supernatant completely.	
	<i>Note: It is of course possible to use larger culture volumes, for example if a large amount of low copy plasmid is needed (see section 4.6 for more information). In this case increase RES, LYS and NEU buffer volumes proportionally in steps 4, 5 and 7. Additional lysis buffer volumes might have to be ordered separately (see ordering information for NucleoBond® Xtra Buffer Set I, section 8.2). Use a centrifuge for the lysate clarification rather than the NucleoBond® Xtra Column Filters.</i>	

Midi	Maxi
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4 Resuspension (Buffer RES)

Resuspend the cell pellet completely in **Resuspension Buffer RES + RNase A** by pipetting the cells up and down or vortexing the cells.

For an efficient cell lysis it is important that no clumps remain in the suspension.

Note: Increase RES buffer volume proportionally if more than the recommended cell mass is used (see section 4.8 for information on optimal cell lysis and section 4.9 regarding difficult-to-lyse strains).

16 mL	24 mL
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5 Cell lysis (Buffer LYS)



Check Lysis Buffer LYS for precipitated SDS prior to use. If a white precipitate is visible, warm the buffer for several minutes at 30–40 °C until precipitate is dissolved completely. Cool buffer down to room temperature (18–25 °C).

Add **Lysis Buffer LYS** to the suspension.

Note: Increase LYS buffer volume proportionally if more than the recommended cell mass is used (see section 4.8 for information on optimal cell lysis).

16 mL	24 mL
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Mix gently by **inverting** the tube **5 times**. **Do not vortex** as this will shear and release contaminating chromosomal DNA from cellular debris into the suspension.

Incubate the mixture at room temperature (18–25 °C) for **5 min**.

Warning: Prolonged exposure to alkaline conditions can irreversibly denature and degrade plasmid DNA and liberate contaminating chromosomal DNA into the lysate.

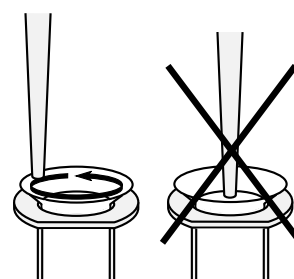
6 Equilibration (Buffer EQU)

Equilibrate a NucleoBond® Xtra Column together with the inserted column filter with **Equilibration Buffer EQU**.

Apply the buffer onto the rim of the column filter as shown in the picture and make sure to wet the entire filter.

Allow the column to empty by gravity flow. The column does not run dry.

12 mL	25 mL
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	Midi	Maxi
7 Neutralization (Buffer NEU)		
	Add Neutralization Buffer NEU to the suspension and immediately mix the lysate gently by inverting the tube until blue samples turns colorless completely! Do not vortex. ! The flask or tube used for this step should not be filled more than two thirds to allow homogeneous mixing. Make sure to neutralize completely to precipitate all the protein and chromosomal DNA. The lysate should turn from a slimy, viscous consistency to a low viscosity, homogeneous suspension of an off-white flocculate. In addition, LyseControl should turn completely colorless without any traces of blue. <i>Note: Increase NEU buffer volume proportionally if more than the recommended cell mass is used (see section 4.8 for information on optimal cell lysis).</i>	
	16 mL	24 mL
	Proceed with step 8 of the high copy plasmid purification protocol (section 7.1) if not significantly more than the recommended lysis buffer volumes were used. Otherwise it might be advantageous to centrifuge the precipitate first in order to avoid clogging of the NucleoBond® Xtra Column Filters.	

7.3 Concentration of NucleoBond® Xtra eluates with NucleoBond® Finalizers

*Note: Use of the **NucleoBond® Finalizers** is only recommended for vector sizes smaller than 50 kbp.*

	Midi – NucleoBond® Finalizer	Maxi – NucleoBond® Finalizer Large
1 Precipitation		
<i>Note: Check DNA concentration photometrically before precipitation. This helps to choose the best buffer volume in step 5 and allows calculation of the recovery after concentration.</i> Add 0.7 volumes of room-temperature isopropanol (not supplied with the kit). Vortex well and let the mixture sit for 2 minutes. (E.g., for 5 mL NucleoBond® Xtra Midi eluate add 3.5 mL isopropanol, for 15 mL NucleoBond® Xtra Maxi eluate add 10.5 mL isopropanol)		
	3.5 mL for 5 mL eluate	10.5 mL for 15 mL eluate
2 Loading		
Remove the plunger from a 30 mL Syringe and attach a NucleoBond® Finalizer to the outlet. Add the precipitation mixture into the syringe, insert the plunger, hold the syringe in a vertical position, and press the mixture slowly through the NucleoBond® Finalizer (the mixture should pass the NucleoBond® Finalizer drop by drop). Discard the flow-through.		
3 Washing		
Remove the NucleoBond® Finalizer from the syringe, pull out the plunger and reattach the NucleoBond® Finalizer to the syringe outlet. Add 70 % ethanol (not supplied with the kit) into the syringe, insert the plunger, hold the syringe in a vertical position, and press the ethanol slowly through the NucleoBond® Finalizer. Discard the flow-through.		
	2 mL	4 mL

**Midi – NucleoBond®
Finalizer**

**Maxi – NucleoBond®
Finalizer Large**

4 Drying

Remove the NucleoBond® Finalizer from the syringe, pull out the plunger and reattach the NucleoBond® Finalizer. Press air through the NucleoBond® Finalizer as strongly as possible while **touching a tissue** with the tip of the NucleoBond® Finalizer to soak up ethanol.

Repeat this step at least as often as indicated below until **no more ethanol** leaks from the NucleoBond® Finalizer.

Note: A new dry syringe can be used to speed up the procedure (not provided).

≥ 6 times until dry

≥ 6 times until dry

Optional: You can incubate the NucleoBond® Finalizer for 10 minutes at 80 °C to minimize ethanol carry-over. However, the final recovery may be reduced by over-drying the DNA.

5 Elution (Buffer TRIS)

Remove the NucleoBond® Finalizer from the syringe, pull out the plunger of a **1 mL Syringe** and attach the NucleoBond® Finalizer to the syringe outlet.

Note: Refer to section 4.13, Table 4 (Midi) or Table 5 (Maxi) to choose the appropriate volume of elution buffer.

Pipette an appropriate volume of **Redissolving Buffer TRIS** (5 mM Tris/HCl, pH 8.5) or TE buffer into the syringe (see section 4.13). Do not use pure water unless pH is definitely higher than 7.0. Place the **NucleoBond® Finalizer** outlet in a vertical position over a fresh collection tube (not provided) and **elute plasmid DNA very slowly** drop by drop by inserting the plunger.

200–800 µL

400–1000 µL

Remove the NucleoBond® Finalizer from the syringe, pull out the plunger and reattach the NucleoBond® Finalizer to the syringe outlet.

Transfer the first eluate back into the syringe and elute into the same collection tube a second time.

Load first eluate completely

Load first eluate completely

For very large expected yields (> 400 µg NucleoBond® Xtra Midi; > 1000 µg NucleoBond® Xtra Maxi recovery can be increased by a third round of elution.

Remove the NucleoBond® Finalizer from the syringe, pull out the plunger to aspirate air, reattach the NucleoBond® Finalizer and **press the air out again to force out as much eluate as possible.**

Determine plasmid yield by UV spectroscopy and confirm plasmid integrity by agarose gel electrophoresis (see section 4.14).

8 Appendix

8.1 Troubleshooting

If you experience problems with reduced yield or purity, it is recommended to check which purification step of the procedure is causing the problem.

First, the bacterial culture has to be checked for sufficient growth (OD_{600}) in the presence of an appropriate selective antibiotic (Table 1, section 4.13). **Second**, aliquots of the cleared lysate, the flow-through, the combined washing steps (Buffer EQU and Buffer WASH), and the eluate should be kept for further analysis by agarose gel electrophoresis.

Refer to Table 6 to choose a fraction volume yielding approximately 5 µg of plasmid DNA assuming 250 µg and 1000 µg were loaded onto the **NucleoBond® Xtra Midi** and **Maxi Column**, respectively. Precipitate the nucleic acids by adding 0.7 volumes of isopropanol, centrifuge the sample, wash the pellet using 70 % ethanol, centrifuge again, remove supernatant, air dry for 10 minutes, dissolve the DNA in 100 µL TE buffer, pH 8.0, and run 20 µL on a 1 % agarose gel.

Table 6: NucleoBond® Xtra eluate volumes required for an analytical check

Sample	Purification step	Volume required [µL]	
		Midi	Maxi
I	Cleared lysate of protocol step 8	500	200
II	Column flow-through after protocol step 8	500	200
III	Wash flow-through after protocol step 9 and 11	250	200
IV	Eluate after protocol step 12	100	100

The exemplary gel picture (Figure 6) will help you to address the specific questions outlined in the following section more quickly and efficiently.

It shows for example the dominant plasmid bands which should only be present in the eluate and in the lysate before loading to proof plasmid production in your cell culture (lane 1). Plasmid DNA found in the wash fractions, however, narrows down the problem to wrong or bad wash buffers (e.g., wrong pH, buffer components precipitated, evaporation of liquid due to wrong storage).

RNA might be visible as a broad band at the bottom of the gel for the lysate and the lysate flow-through samples (lanes 1 and 2). It might also occur in the wash fraction but must be absent in the eluate.

Genomic DNA should not be visible at all but would show up in the gel slot or right below indicating for example too harsh lysis conditions.

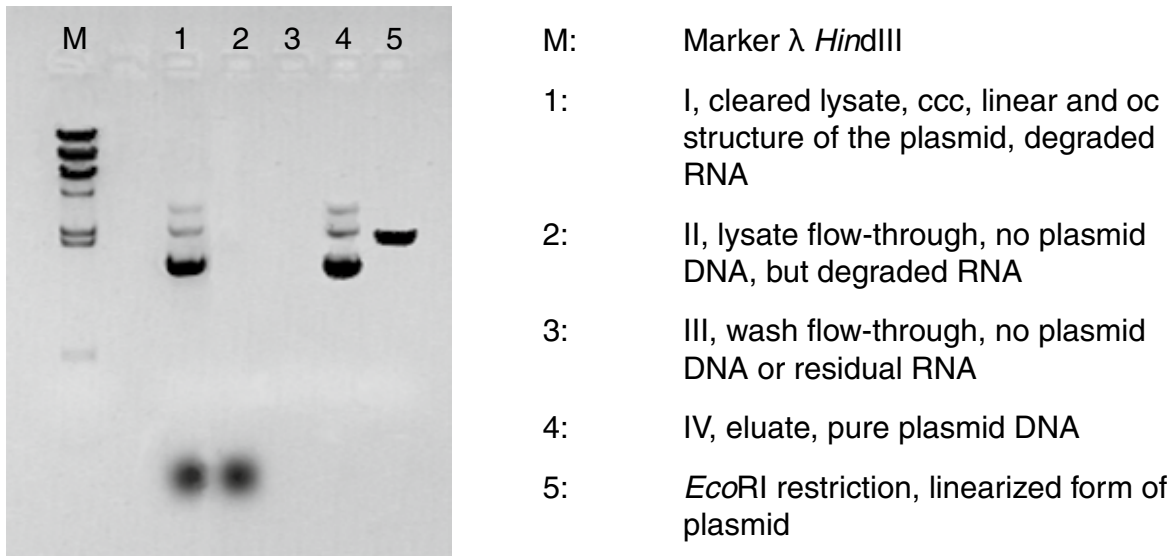


Figure 6 Exemplary analytical check of NucleoBond® Xtra Midi purification samples
Plasmid: pUC18, bacterial strain: *E. coli* DH5 α ®. 20 μ L of each precipitated sample has been analyzed on a 1 % agarose gel. Equal amounts of plasmid DNA before (lane 1) and after (lane 4) purification using NucleoBond® Xtra Midi are shown with a recovery of > 90 %.

Problem	Possible cause and suggestions
No or low plasmid DNA yield	<i>Plasmid did not propagate</i> - Check plasmid content in the cleared lysate (see Figure 6). Use colonies from fresh plates for inoculation and add fresh selective antibiotic to plates and media. - Estimate plasmid content prior to large purifications by a quick NucleoSpin® Plasmid or NucleoSpin® Plasmid EasyPure preparation.
	<i>Alkaline lysis was inefficient</i> - Too much cell mass was used. Refer to section 4.5–4.8 regarding recommended culture volumes and lysis buffer volumes. Check plasmid content in the cleared lysate (see Figure 6). - Check Buffer LYS for SDS precipitation before use, especially after storage below 20 °C. If necessary incubate the bottle for several minutes at 30–40 °C and mix well until SDS is redissolved.
	<i>SDS- or other precipitates are present in the sample</i> Load the crude lysate onto the NucleoBond® Xtra Column Filter inserted in the NucleoBond® Xtra Column. This ensures complete removal of SDS precipitates. Incubation of cleared lysates for longer periods of time might lead to formation of new precipitate. If precipitate is visible, it is recommended to filter and centrifuge the lysate again directly before loading it onto the NucleoBond® Xtra Column.
	<i>Sample/lysate is too viscous</i> - Too much cell mass was used. Refer to section 4.5–4.8 regarding recommended culture volumes and lysis buffer volumes. - Make sure to mix well after neutralization to completely precipitate SDS and chromosomal DNA. Otherwise, filtration efficiency and flow rate go down and SDS prevents DNA from binding to the column.
	<i>pH or salt concentrations of buffers are too high</i> Check plasmid content in the wash fractions (see Figure 6). Keep all buffers tightly closed. Check and adjust pH of Buffer EQU (pH 6.5), WASH (pH 7.0), and ELU (pH 9.0) with HCl or NaOH if necessary.

Problem	Possible cause and suggestions
NucleoBond® Xtra Column Filter clogs during filtration	<i>Culture volumes are too large</i>
	Refer to section 4.5–4.8 regarding recommended culture volumes and larger lysis buffer volumes.
	<i>Precipitate was not resuspended before loading</i> Invert crude lysate at least 3 times directly before loading.
NucleoBond® Xtra Column is blocked or very slow	<i>Incomplete precipitation step</i>
	Make sure to mix well after neutralization to completely precipitate SDS and chromosomal DNA.
	<i>Sample is too viscous</i>
NucleoBond® Xtra Column Filter or centrifuge at higher speed or for a longer period of time.	Do NOT attempt to purify lysate prepared from a culture volume larger than recommended for any given column size with standard lysis buffer volumes. Incomplete lysis not only blocks the column but can also significantly reduce yields. Refer to section 4.5 and 4.6 for recommended culture volumes and section 4.8 for larger culture volumes and adjusted lysis buffer volumes.
	Make sure to mix well after neutralization to completely precipitate SDS and chromosomal DNA.
	<i>Lysate was not cleared completely</i>
Genomic DNA contamination of plasmid DNA	Use NucleoBond® Xtra Column Filter or centrifuge at higher speed or for a longer period of time.
	Precipitates occur during storage. Clear lysate again before loading the column.
	<i>Lysis treatment was too harsh</i>
Genomic DNA contamination of plasmid DNA	Make sure not to lyse in Buffer LYS for more than 5 min.
	<i>Lysate was mixed too vigorously or vortexed after lysis</i>
	- Invert tube for only 5 times. Do not vortex after addition of Buffer LYS. - Use larger tubes or reduce culture volumes for easier mixing.

Problem	Possible cause and suggestions
RNA contamination of plasmid DNA	<i>RNase digestion was inefficient</i> RNase was not added to Buffer RES or stored improperly. Add new RNase to Buffer RES. See section 8.2 for ordering information.
	<i>pH or salt concentration of wash buffer is too low</i> - Check RNA content in the wash fractions (see Figure 6). Keep all buffers tightly closed. Check pH of Buffer EQU (pH 6.5) and WASH (pH 7.0) and adjust with HCl or NaOH if necessary. - Increase wash buffer stringency by adjusting pH of Buffer WASH to 7.5.
	<i>Wash step with Buffer WASH was not sufficient</i> Double or triple washing step with Buffer WASH. Additional Buffer WASH can be ordered separately (see ordering information).
Low purity ($A_{260}/A_{280} < 1.8$)	<i>NucleoBond® Xtra Column Filter was not removed before second washing step</i> Protein content too high due to inefficient washing. Remove the NucleoBond® Xtra Column Filter before performing the second washing step with Buffer WASH.
	<i>Buffer WASH was used instead of Buffer EQU for the first wash</i> Buffer EQU has to be used to wash out the NucleoBond® Xtra Column Filter to avoid SDS carry-over.
	<i>Only minimal amounts of DNA were loaded onto the column</i> - Excess free binding capacity requires more extensive washing – double washing step with Buffer WASH. - Reduce lysis time < 5 min.
No nucleic acid pellet formed after precipitation	<i>Pellet was lost</i> Handle the precipitate with care. Decant solutions carefully. Determine DNA yield in Buffer ELU in order to calculate the amount of plasmid DNA that should be recovered after precipitation.
	<i>Plasmid DNA might be smeared over the wall of the tube</i> Dissolve DNA with an appropriate volume of reconstitution buffer by rolling the tube for at least 30 min.
	<i>Nucleic acid did not precipitate</i> - Check type and volumes of precipitating solvent. Make sure to use at least 0.7 volumes of isopropanol and mix thoroughly. - Centrifuge for longer periods of time at higher speed.

Problem	Possible cause and suggestions
Nucleic acid pellet is opaque or white instead of clear and glassy	<i>Co-precipitation of salt</i>
	- Check isopropanol purity, and perform precipitation at room temperature (18–25 °C) but centrifuge at 4 °C. Do not let the eluate drip from the column into isopropanol but add isopropanol to the final eluate and mix immediately. - Try resuspending the pellet in Buffer WASH, and reload onto the same NucleoBond® Xtra Column. Wash the column several times with Buffer WASH before loading.
Nucleic acid pellet does not resuspend in buffer	<i>Pellet was over-dried</i>
	Try to dissolve at higher temperatures for a longer period of time (e.g., 2 h at 37 °C or overnight at RT), preferably under constant spinning (3D-shaker).
	<i>Co-precipitation of salt or residual alcohol</i>
Nucleic acid pellet does not resuspend in buffer	Wash the pellet again with 70 % ethanol, or increase the reconstitution buffer volume.
	<i>Insoluble particles in redissolved DNA</i>
	Centrifuge the redissolved DNA to pellet the insoluble particles and transfer supernatant to a new tube. Insoluble particles do not affect DNA quality. As an alternative insoluble particles can easily be removed by using the NucleoBond® Finalizer (NucleoBond® Xtra Midi) or NucleoBond® Finalizer Large (NucleoBond® Xtra Maxi).

Problem	Possible cause and suggestions
No or low plasmid DNA yield after NucleoBond® Finalizer precipitation	<i>Already no or low plasmid DNA after elution from the NucleoBond® Xtra Column</i>
	Refer to detailed troubleshooting “No or low plasmid DNA yield”.
	<i>Dead volume too high</i>
	If high concentration of plasmid DNA is the main priority, elution should be performed in small volumes. Naturally a portion of the eluate will be lost in the syringe and on the NucleoBond® Finalizer. To minimize these losses in the second elution step, try to transfer even the last droplet from the syringe to the NucleoBond® Finalizer, for example by tapping the NucleoBond® Finalizer and syringe onto the bench top. Then fill the syringe with air and press forcefully the last droplets out of the NucleoBond® Finalizer. Repeat this step several times. You might have to practice this procedure several times to achieve optimal results. An acceptable dead volume is smaller than 30 µL with NucleoBond® Finalizer and 60 µL with NucleoBond® Finalizer Large.
	<i>Elution volume too small</i>
	Since there are dead volumes of about 30 µL (NucleoBond® Finalizer) and 60 µL (NucleoBond® Finalizer Large), reasonable elution volumes start with 200 µL (NucleoBond® Finalizer) and 400 µL (NucleoBond® Finalizer Large) respectively. Further-more, smaller volumes are insufficient to wet the entire membrane and will drastically decrease your yield. Refer to section 4.13, Table 4 and Table 5 to estimate the recovery that can be expected depending on elution buffer volume.
	<i>Elution too fast</i>
	Plasmid DNA needs time to dissolve. Elute really very slowly, drop by drop. Repeat the elution procedure using the first eluate.
	<i>Forgot to elute a second time</i>
	Repeating the elution procedure with the first eluate is crucial for optimal yields. However, eluting a third time shows no further improvement.
	<i>Plasmid size</i>
	Precipitation efficiency is almost independent of plasmid size, but elution from the NucleoBond® Finalizers becomes more and more difficult with increasing size of the construct. If you face low yields with large cosmids you may try heating the NucleoBond® Finalizer, the syringes, and elution buffer to 70 °C.

Problem	Possible cause and suggestions
Low DNA concentration after NucleoBond® Finalizer precipitation	<i>Low overall yield</i>
	Refer to detailed troubleshooting “No or low plasmid DNA yield” and lower your elution buffer volume. Refer to section 4.13, Table 4 and Table 5 to estimate the DNA concentrations that can be expected.
	<i>Fresh elution buffer was used for second elution step</i>
Low recovery when using the Finalizer	The second elution step is crucial for optimal yield but to achieve a high DNA concentration the eluate of the first elution step has to be used for the second elution.
	<i>Not enough DNA loaded</i>
	Since there is a technical limitation to at least 200 µL (NucleoBond® Finalizer) and 400 µL (NucleoBond® Finalizer Large) of elution buffer due to membrane wetting and dead volume, a minimal amount of DNA has to be loaded to achieve a desired concentration. If possible try to pool several DNA precipitation batches since percentage of recovery and concentration significantly increase with higher amounts of loaded DNA.
Low recovery when using the Finalizer	<i>Resuspension time might be too short</i>
	Depending on the total amount of the precipitated plasmid it will need some time to redissolve completely. Resuspension time might be too short for a complete recovery, if the redissolving buffer is passed by the precipitated plasmid on the Finalizer membrane too quickly.
	If a high recovery is mandatory it is recommended to incubate the precipitated plasmid in the redissolving buffer during the elution step. Therefore, do not press the elution buffer through the Finalizer in one rush, but rather stop pressing the buffer through as soon as the first drops have passed the outlet. Let the redissolving buffer sit on the membrane in the Finalizer for 5 minutes at room temperature before completing the elution step. Reload the eluate onto the Finalizer and repeat the procedure at least once. General recommendations are also valid here: push the redissolving buffer through the membrane slowly, increase the elution volume to gain a higher recovery and blow out the dead volume completely by pushing air through the Finalizer.

Problem	Possible cause and suggestions
Purified plasmid does not perform well in subsequent reactions	<i>Plasmid DNA is contaminated with chromosomal DNA or RNA</i>
	Refer to the detailed troubleshooting above.
	<i>Plasmid DNA is contaminated with residual alcohol</i>
	Plasmid DNA was not dried completely before redissolving. Precipitate DNA again by adding 1/10 volume of 3 M NaAc pH 5.0 and 0.7 volumes of isopropanol. Proceed with the precipitation protocol in this manual and dry DNA pellet completely.
	<i>DNA is degraded</i>
	- Make sure that your entire equipment (pipettes, centrifuge tubes, etc.) is clean and nuclease-free. - Do not lyse the sample with Buffer LYS for more than 5 min.
	<i>DNA is irreversibly denatured</i>
	A denatured plasmid band runs faster on the gel than the supercoiled conformation. Do not lyse the sample after addition of Buffer LYS for more than 5 minutes.

8.2 Ordering information

Product	REF	Pack of
NucleoBond® Xtra Midi	740410.10/.50/.100	10/50/100 preps
NucleoBond® Xtra Midi Plus (incl. NucleoBond® Finalizers)	740412.10/.50	10/50 preps
NucleoBond® Xtra Maxi	740414.10/.50/.100	10/50/100 preps
NucleoBond® Xtra Maxi Plus (incl. NucleoBond® Finalizers Large)	740416.10/.50	10/50 preps
NucleoBond® Xtra Combi Rack	740415	1
NucleoBond® Xtra Buffer Set I (Buffer RES, LYS (with LyseControl), NEU, RNase A; only applicable with NucleoBond® Xtra kits; sufficient for 12 Xtra Maxi and 18 Xtra Midi preps)	740417	1
Buffer RES	740363.1000	1000 mL
Buffer EQU	740317.1000	1000 mL
Buffer WASH	740375.1000	1000 mL

Product	REF	Pack of
Buffer ELU	740316.600	600 mL
NucleoBond® Finalizer (for use with NucleoBond® Xtra Midi, Midi EF, NucleoBond® PC 100, PC 500, PC 500 EF)	740519.20	20 filters 2 syringe sets
NucleoBond® Finalizer Plus (for use with NucleoBond® Xtra Midi, Midi EF, NucleoBond® PC 100, PC 500, PC 500 EF)	740520.20	20 filters 20 syringe sets
NucleoBond® Finalizer Large (for use with NucleoBond® Xtra Maxi, Maxi EF, NucleoBond® PC 2000, PC 2000 EF)	740418.20	20 large filters 2 syringe sets
NucleoBond® Finalizer Large Plus (for use with NucleoBond® Xtra Maxi, Maxi EF, NucleoBond® PC 2000, PC 2000 EF)	740419.20	20 large filters 20 syringe sets
RNase A (lyophilized)	740505.50 740505	50 mg 100 mg
NucleoMag® Desalting Beads (for desalting of NucleoBond® eluates)	744410.50	4 x 1.5 mL and Buffer TRIS (60 mL)

Visit www.mn-net.com for more detailed product information.

8.3 Product use restriction / warranty

NucleoBond® Xtra Midi / Maxi kit components are intended, developed, designed, and sold FOR RESEARCH PURPOSES ONLY, except, however, any other function of the product being expressly described in original MACHEREY-NAGEL product leaflets.

MACHEREY-NAGEL products are intended for GENERAL LABORATORY USE ONLY! MACHEREY-NAGEL products are suited for QUALIFIED PERSONNEL ONLY! MACHEREY-NAGEL products shall in any event only be used wearing adequate PROTECTIVE CLOTHING. For detailed information please refer to the respective Material Safety Data Sheet of the product! MACHEREY-NAGEL products shall exclusively be used in an ADEQUATE TEST ENVIRONMENT. MACHEREY-NAGEL does not assume any responsibility for damages due to improper application of our products in other fields of application. Application on the human body is STRICTLY FORBIDDEN. The respective user is liable for any and all damages resulting from such application.

DNA/RNA/PROTEIN purification products of MACHEREY-NAGEL are suitable for IN-VITRO-USES ONLY!

ONLY MACHEREY-NAGEL products specially labeled as IVD are also suitable for IN-VITRO-diagnostic use. Please pay attention to the package of the product. IN-VITRO-diagnostic products are expressly marked as IVD on the packaging.

IF THERE IS NO IVD SIGN, THE PRODUCT SHALL NOT BE SUITABLE FOR IN-VITRO-DIAGNOSTIC USE!

ALL OTHER PRODUCTS NOT LABELED AS IVD ARE NOT SUITED FOR ANY CLINICAL USE (INCLUDING, BUT NOT LIMITED TO DIAGNOSTIC, THERAPEUTIC AND/OR PROGNOSTIC USE).

No claim or representations is intended for its use to identify any specific organism or for clinical use (included, but not limited to diagnostic, prognostic, therapeutic, or blood banking). It is rather in the responsibility of the user or - in any case of resale of the products - in the responsibility of the reseller to inspect and assure the use of the DNA/RNA/protein purification products of MACHEREY-NAGEL for a well-defined and specific application.

MACHEREY-NAGEL shall only be responsible for the product specifications and the performance range of MN products according to the specifications of in-house quality control, product documentation and marketing material.

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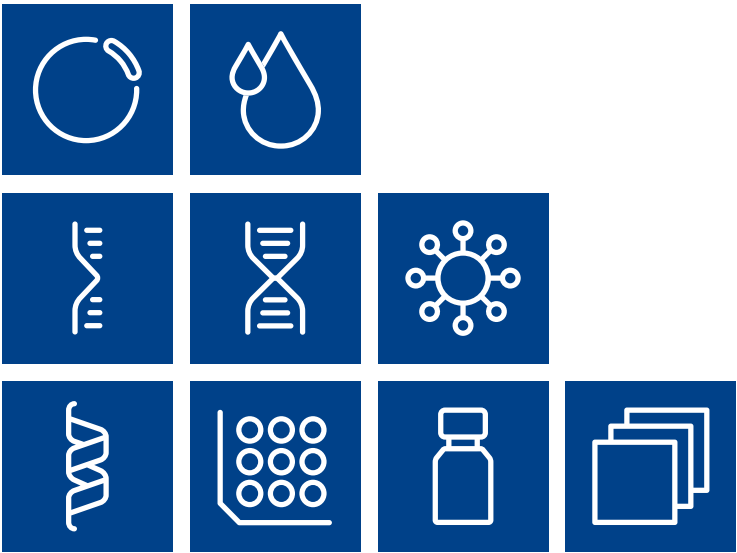
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Plasmid DNA

Clean up

RNA

DNA

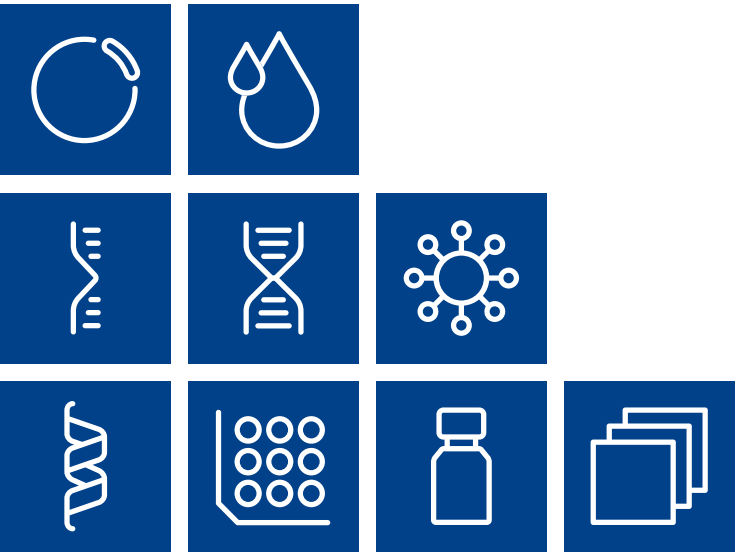
Viral RNA and DNA

Protein

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Accessories

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